MILITARY SPECIFICATION

CONNECTOR, SWITCH

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch and connector assembly entitled as 'Selector Switch and Connector Assembly which is used in the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8800543	- Carton, Packing, Ammunition, For Selector Switch and Connector Assembly.
8800814	- Switch, Selector Assembly.
8808362	- Container, Packing, Ammunition, For Domestic (CONUS) Shipments of Special Weapons Items. Packing and Marking.
8831043	- Selector Switch and Connector Assembly.
EL-8842567	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The selector switch shall comply with all requirements specified on the drawing (dwg) 8831043, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Circuit resistance. With the selector switch in the position indicated in Table I, the resistance of each continuity circuit indicated for the selector switch position shall not exceed the value specified in Table I when determined as specified in 4.4.1.

Table I

Circuit resistance

Test No.	Selector Switch Position	From Switch (S2) Terminal	To S2 or 5J2 Terminal	Maximum Resistance Ohms
1	1	A-33 (Red lead)*	5J2-V	0.2
2	1	B-33 (Brown lead)*	5J2-P	0.2
3	1	E-33 (Green lead)*	D-10 (Black lead)*	0.2
4	2	A-33	5J2-$\overline{t}$	0.2
5	2	B-33	5J2-$\overline{x}$	0.2
6	2	E-33	D-10	0.2
7	2	G-33	G-2	0.2
8	3	A-33	5J2-FF	0.2
9	3	B-33	5J2-GG	0.2
10	3	E-33	D-10	0.2
11	4	A-33	5J2-EE	0.2
12	4	B-33	5J2-$\overline{m}$	0.2
13	4	E-33	D-10	0.2
14	5	A-33	5J2-$\overline{q}$	0.2
15	5	B-33	5J2-$\overline{k}$	0.2
16	5	E-33	D-10	0.2
17	6	A-33	5J2-$\overline{r}$	0.2
18	6	B-33	5J2-$\overline{I}$	0.2
19	6	E-33	D-10	0.2
20	7	A-33	5J2-$\overline{r}$	0.2
21	7	B-33	5J2-$\overline{h}$	0.2
22	7	E-33	D-10	0.2
23	8	A-33	5J2-$\overline{e}$	0.2
24	8	B-33	5J2-$\overline{a}$	0.2
25	8	E-33	D-10	0.2

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Test No.	Selector Switch Position	From Switch (S2) Terminal	To S2 or 5J2 Terminal	Maximum Resistance Ohms
26	9	A-33	5J2-Y	0.2
27	9	B-33	5J2-Z	0.2
28	9	E-33	D-10	0.2
29	10	A-33	5J2-d	0.2
30	10	B-33	5J2-n	0.2
31	10	E-33	D-10	0.2
32	11	A-33	5J2-Z	0.2
33	11	B-33	5J2-f	0.2
34	11	E-33	D-10	0.2
35	12	A-33	5J2-d	0.2
36	12	B-33	5J2-Y	0.2
37	12	E-33	D-10	0.2
38	13	A-33	5J2-X	0.2
39	13	B-33	5J2-HH	0.2
40	13	E-33	D-10	0.2
41	14	A-33	5J2-V	0.2
42	14	B-33	5J2-R	0.2
43	14	E-33	D-10	0.2
44	15	A-33	5J2-J	0.2
45	15	B-33	5J2-H	0.2
46	15	E-33	D-10	0.2
47	16	A-33	5J2-J	0.2
48	16	B-33	5J2-G	0.2
49	16	E-33	D-10	0.2
50	17	A-33	5J2-DD	0.2
51	17	B-33	5J2-CC	0.2
52	17	E-33	D-10	0.2
53	18	A-33	5J2-EE	0.2
54	18	B-33	5J2-K	0.2
55	18	E-33	D-10	0.2
56	19	B-33	5J2-X	0.2
57	19	H-33 (Brown lead)*	5J2-e	0.2
58	19	E-33	5J2-Z	0.2
59	19	E-33	F-29 (White-black)*	0.2
60	20	A-33	5J2-q	0.2
61	20	B-33	5J2-p	0.2
62	20	E-33	D-10	0.2
63	21	B-33	5J2-K	0.2
64	21	E-33	5J2-J	0.2
65	21	E-33	F-29	0.2
66	22	B-33	5J2-p	0.2
67	22	H-33	5J2-d	0.2
68	22	E-33	5J2-q	0.2
69	22	E-33	F-29	0.2
70	23	B-33	5J2-a	0.2
71	23	H-33	5J2-p	0.2
72	23	E-33	5J2-EE	0.2
73	23	E-33	F-29	0.2
74	24	B-33	5J2-EE	0.2
75	24	E-33	5J2-q	0.2
76	24	E-33	F-24 (White-green)*	0.2
77	24	G-33 (Violet)*	D-10 (Black)*	0.2

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Test No.	Selector Switch Position	From Switch (S2) Terminal	To S2 or SJ2 Terminal	Maximum Resistance Ohms
78	25	B-33	5J2-h	0.2
79	25	H-33	5J2-d	0.2
80	25	E-33	5J2-EE	0.2
81	25	E-33	F-29	0.2
82	26	B-33	5J2-FF	0.2
83	26	H-33	5J2-W	0.2
84	26	E-33	5J2-X	0.2
85	26	E-33	F-29	0.2
86	27	A-33	5J2-EE	0.2
87	27	B-33	5J2-F	0.2
88	27	E-33	D-10	0.2
89	28	B-33	5J2-X	0.2
90	28	H-33	5J2-W	0.2
91	28	E-33	5J2-V	0.2
92	28	E-33	F-29	0.2
93	29	B-33	5J2-x	0.2
94	29	H-33	5J2-w	0.2
95	29	5J2-EE	5J2-V	0.2
96	29	E-33	F-29	0.2
97	30	A-33	5J2-EE	0.2
98	30	B-33	5J2-C	0.2
99	30	E-33	D-10	0.2
100	31	A-33	5J2-s	0.2
101	31	B-33	5J2-w	0.2
102	31	E-33	D-10	0.2
103	31	G-33	G-2 (Slate)*	0.2
104	32	A-33	5J2-t	0.2
105	32	B-33	5J2-w	0.2
106	32	E-33	D-10	0.2
107	32	G-33	G-2	0.2

*The color of these leads are for reference only and are not mandatory.

3.5 Insulation resistance. With the selector switch in position 21 and when checked at 500 volts ± 50 volts direct current (vdc) the resistance of each terminal to all other terminals listed in Table II shall be greater than 100 megohms. With the selector switch in position 1, the resistance of 5J2-J to 5J2-K shall be greater than 100 megohms when checked at 500 ± 50 vdc. Testing shall be as specified in 4.4.2.

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Table II

Insulation resistance

5J2 Connector Terminals

			Unused Terminals (note 1)
B	a	s	D
C	d	t	E
F	e	w	L
G	f	x	M
H	g	y	N
J	h	z	S
P	i	AA	T
R	k	CC	U
V	m	DD	b
W	n	EE	c
X	p	FF	j
Y	q	GG	u
Z	r	HH	Connector shell - note 2
Y	BB		Metal frame of switch - note 2

Note 1: Unused terminals shall be connected together for this test and tested as one terminal.
Note 2: Connector shell and metal switch frame shall be at common ground potential which will also be test fixture ground.

3.6 Selector switch mechanism. The selector switch shall be capable of being rotated only one position at a time in the clockwise direction and no positions in the counter-clockwise direction. The switch shall not bind in any position. Testing shall be as specified in 4.4.3.

3.7 Environmental testing. The selector switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The selector switch shall also comply with applicable paragraphs in this section before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.7.1 Transportation vibration. The selector switch and connector assembly shall be vibrated over the complete frequency range of Table III for one cycle in the direction of the switch shaft and for one cycle in a direction perpendicular to the shaft at the frequencies, amplitudes, and times specified in Table III without evidence of damage such as cracks, chips, broken leadwires, separations or loosening parts. The switch and connector assembly shall comply with the requirements of 3.4 and 3.5 after vibration. Testing shall be as specified in 4.4.4.1.

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Table III

Transportation vibration

Test Cycle Hertz (10% tol.)	Cycle Time to Cover Frequency Range (minutes)	Double Amplitude or Peak g Level (+ 5% to 1)	Total Cycling Time in Minutes
10-20-10*	3 ± 0.5	± 2 g's	15 minimum (min)
20-500-20*	3 ± 0.5	± 5 g's	15 min

*Either a uniform or a logarithmic rate of change of frequency over the test cycle shall be maintained.

3.7.2 Transportation shock. The selector switch and connector assembly shall withstand six 20 gravity (g) shocks, two in each direction of the switch shaft and two in a direction perpendicular to the shaft, without evidence of damage such as cracks, chips, broken leadwires, separations, or loosening of parts. The assembly shall comply with the requirements of 3.4 and 3.5 after shocking. Testing shall be as specified in 4.4.4.2.

3.7.3 High temperature. The assembly shall withstand a temperature of 160 plus or minus (±) 5 degrees Fahrenheit (°F) without damage or deterioration and the assembly shall comply with 3.6 while at 160 ± 5°F when tested as specified in 4.4.4.3.

3.7.4 Low temperature. The assembly shall withstand a temperature of -80 ± 5°F without evidence of damage or deterioration and the assembly shall comply with 3.6 while at -65 ± 5°F when tested as specified in 4.4.4.4.

3.7.5 Temperature shock. The selector switch and connector assembly shall withstand temperature shock of -65 ± 5°F to 80 ± 5°F to 160 ± 5°F without evidence of cracking, peeling, swelling or other damage when tested as specified in 4.4.4.5. Upon completion of this test the assembly shall comply with 3.4 and 3.5.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

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3.8.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8831043) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Circuit resistance (see 3.4)		X
3. Insulation resistance (see 3.5)		X
4. Selector switch mechanism (see 3.6)		X
5. Transportation vibration (see 3.7.1)	X	
6. Transportation shock (see 3.7.2)	X	
7. High temperature (see 3.7.3)		X
8. Low temperature (see 3.7.4)		X
9. Temperature shock (see 3.7.5)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch, selector assembly (see dwg 8800814 covering a detail of dwg 8831043).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect retard mechanism adjustment	Gage
	102. Gear not pinned to switch shaft	Visual
	103. Any part missing or damaged to extent that functioning is impaired	Visual
	104. Any misaligned, loosely connected or mislocated component part(s)	Visual/Manual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Selector switch and connector assembly (see dwg 8831043).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Permanently damaged connector terminals	Visual

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102.	Any connector socket terminals missing (as evidenced by empty hole in insert)	Visual
103.	Portion of connector broken away (minor chips, scratches or gouges which do not affect mating are not objectionable)	Visual
104.	Voids in monitor plug potting greater than 1/8 inch in diameter	Visual
105.	Wrong connector	Visual
*106.	Depth of sockets (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent sockets)	Visual
107.	Wire nicked or broken	Visual
108.	Evidence of poor workmanship (see 3.8)	Visual
109.	Evidence of poor solder joint (see dwg 8831043)	Visual
110.	Any missing, misaligned, loosely connected or mislocated component parts	Visual/Manual
111.	Mechanism prevents movable switch from aligning on fixed switch contacts	Visual/Manual
112.	Switch binds and will not release	Visual/Manual
113.	Switch can be rotated in counter-clockwise direction	Manual
114.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor:

201.	Stamping missing or incorrect	Visual
202.	Sleeving on wires and terminals loose	Visual/Manual
203.	Cable binding ties loose or producing tension on soldered connection	Visual/Manual

*In the event socket inserts do not appear to be within 1/16 inch in depth relative to adjacent sockets, a gage (see 4.3.4.1) shall be used to determine that variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward edge of the connector shall be utilized as a reference surface for all measurements of socket depths.

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4.3.2.1.3 Carton packing ammunition, for selector switch and connector assembly (see dwg 8800543).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Packing material visibly wet	Visual
	102. Incorrect or insufficient cushioning material	Visual
	103. Evidence of poor workmanship (see 3.8)	Visual
	104. Wrong size bag	Visual
	105. Bag not sealed	Visual
	106. Air in bag not minimized before sealing	Visual
	107. Marking missing, mislocated or incorrect	Visual
Minor:	None defined.	

4.3.2.1.4 Container, packing, ammunition for domestic (CONUS) shipments of special weapon items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Insufficient cushioning material	Visual
	102. Marking missing or incorrect	Visual
	103. Evidence of poor workmanship (see 3.8)	Visual
	104. Packing material visibly wet	Visual
Minor:	None defined.	

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Circuit resistance	Major
b.	Insulation resistance	Major
c.	Selector switch mechanism	Major
d.	Transportation vibration	Major

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Of these tests, d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance, insulation resistance and selector switch mechanism. Each selector switch assembly in the lot shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.4, 3.5 and 3.6, the lot shall be rejected and removed from the lot.

4.3.3.2 Transportation vibration. A sample shall be selected at random in accordance with Table VII. The sample shall be tested in accordance with 4.4.4.1. If any item fails to comply with 3.7.1, the lot shall be rejected.

Table VII

Sample sizes for destructive testing

Number Required to be Delivered	Sample Size for No Failures
2-180	2
181-500	3
501-800	5
801-1300	7

Lot Size Required = Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842567). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, testing shall be performed at 80 ± 20°F.

4.4.1 Circuit resistance. The circuit resistance between the terminals specified in Table I shall be determined with the use of a suitable electrical resistance measuring instrument.

4.4.2 Insulation resistance. The insulation resistance between the terminals specified in Table II shall be determined with the use of an insulation resistance tester, capable of supplying the required voltage and from which a reading directly in megohms can be made. The current shall be limited to a maximum of 5 milliamperes to protect human life.

4.4.3 Selector switch mechanism. The selector switch and connector assembly shall be positioned such that its major axis is in a vertical direction. Turn the test selector switch at a medium to fast rate of speed through all 32 switch positions without exerting undue force at any time. Apply a slow slight counterclockwise force to the switch after reaching each position to allow advancing to the next position and to check for prevention of counterclockwise rotation.

4.4.4 Environmental testing (see 3.7).

4.4.4.1 Transportation vibration. The selector switch and connector assembly shall be fastened securely on a suitable vibration machine such as to permit vibration in the direction of the switch shaft and in the direction perpendicular to the shaft. The test mounting shall simulate the mounting of the assembly in the Storage Tester. The assembly shall be vibrated at the frequencies, amplitudes, and for the times specified in Table III in the directions specified. Following this test, the assembly shall be tested as specified in 4.4.1 and 4.4.2.

4.4.4.2 Transportation shock. The selector switch and connector assembly shall be mounted to a test fixture to simulate the mounting of the assembly in the Test Set T-4087 Chassis Assembly. The fixture shall then be mounted to a shock producing machine and the switch subjected to 2 shocks in each of the specified directions for a total of 6 shocks. These shocks shall be 20 ± 2 g shocks of 11 ± 1 milliseconds half amplitude duration. Following this test, the assembly shall be tested as specified in 4.4.4.1 and 4.4.4.2.

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4.4.4.3 High temperature. The switch and connector assembly shall be placed within the chamber and the internal temperature of the chamber raised to 160 + 5°F and stabilized (see 6.4). The switch and connector assembly shall be maintained at 160 + 5°F for a period of 2 hours minimum. At the conclusion of this exposure period and while still at this temperature, the assembly shall be operated as specified in 4.4.3.

4.4.4.4 Low temperature. The switch and connector assembly shall be enclosed to give it approximately the same degree of frost protection it would have when mounted in a Test Set, T-4087. It shall then be placed within a chamber and the chamber cooled to and maintained at a temperature of -80 + 5°F for a period of 4 hours minimum. The temperature of the chamber shall then be raised to -65 + 5°F and maintained at this temperature for an additional 2 hour period. At the conclusion of this exposure period and while still at this temperature, the assembly shall be operated as specified in 4.4.3.

4.4.4.5 Temperature shock. The selector switch and connector assembly shall be placed within a chamber having a temperature of -65 + 5°F. The assembly shall be subjected to this temperature for a period of 4 hours minimum, at the conclusion of which, and within 5 minutes, the switch and connector assembly shall be transferred to a chamber having an internal temperature of 80 + 20°F. The assembly shall be subjected to this temperature for a minimum period of 4 hours, after which it shall be transferred, within 5 minutes, to a test chamber having an internal temperature of 160 + 5°F. The assembly shall be subjected to this temperature for a minimum period of 4 hours. At the conclusion of this exposure and within 5 minutes, the assembly shall be transferred to the -65 + 5°F test chamber and shall be subjected to this temperature for a minimum of 4 hours. This completes one cycle. The number of complete cycles shall be three. At the conclusion of the third cycle, the assembly shall be removed from the test chamber and tested as specified in 4.4.1 and 4.4.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The selector switch and connector assembly shall be packaged in accordance with dwg 8800543 using the specified moisture barrier.

5.1.2 Level C. The selector switch and connector assembly shall be packaged in accordance with dwg 8800543 except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The selector switch and connector assembly shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8800543 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1691 dated 6 October 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A014

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